

ABHAY BHANDARKAR

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Portfolio: abhaybhandarkar.github.io

Github ◊ LinkedIn

OBJECTIVE & RESEARCH INTERESTS

Fourth-year undergraduate researcher bridging perception and generation. Deeply interested in Multi-modal AI, Computer Vision, and Applied Machine Learning, with a specific focus on visual reasoning, representation learning, and generative modeling for content creation. Seeking PhD opportunities in Machine Learning and Computer Vision.

EDUCATION

Ramaiah Institute of Technology (RIT)

Bachelor of Engineering in Computer Science and Engineering

Expected September 2026

CGPA: 8.06/10.0

TECHNICAL SKILLS

Programming Languages

Python, C/C++

ML/DL Frameworks

PyTorch, PyTorch Lightning, TensorFlow, HuggingFace

Tools & Platforms

Git, Docker, Linux, Roboflow, L^AT_EX

RESEARCH EXPERIENCE

Research Intern

Jan 2026 - Present

Advisor: Dr. Vineeth N Balasubramanian

Microsoft Research India

- Spearheading foundational research in generative modeling, focusing on the theoretical underpinnings and practical applications of Diffusion Models.
- Developed *Pixels to Layers*, an architecture turning generated infographics into editable assets, resulting in an Oral presentation and Demo acceptance at **CVPR 2026**.
- Engineered *LEGO* (Learned Editable Graphic Outputs for Infographic Decomposition), an applied framework for adaptive infographic decomposition (**WACV 2027** under review).
- Co-authored papers currently under submission at **WACV 2027** and have a patent in the process of grant.

ACM Summer Uplink Intern

July 2025 - Dec 2025

Advisor: Dr. Vineeth N Balasubramanian

ACM IKDD

- Conducted foundational research in generative modeling with a focus on Diffusion Models.
- Implemented and experimented with state-of-the-art diffusion architectures for image decomposition and editing tasks.
- Appointed as a Business Guest at Microsoft Research Bangalore to contribute to MicrosoftTools.

Samsung PRISM Elite Intern

Sep, 2025 - Apr, 2026

- Developing Distributed LoRA systems to enhance parameter-efficient fine-tuning for large-scale models.

AI Research Fellow

May 2025 - Aug 2025

Remote

Pradhi AI Solutions, India

- Conducted advanced research into scalable AI architectures and novel machine learning models in voice intelligence.
- Collaborated with a team of senior developers to write production-grade code with meticulous testing.

Data Science Intern

Product Team

Sep 2024 - Nov 2024

Altqube, India

- Developed a YOLOv8 model for the detection of Lumpy Skin Disease in cattle with 96% accuracy.
- Devised a pipeline for Oriented Bounding Box format to analyze rice grain quality in an industrial setting.
- Implemented a pipeline for training medical images with the Nomic AI Vision Embedding model.
- Annotated 600+ images with multi-label bounding boxes on Roboflow to facilitate robust YOLOv8 training.

Machine Learning Intern

Karnataka Government Transliteration Project

2023 - 2024

Crisant Technologies

- Assisted with English-to-Kannada transliteration for streamlined documentation for the Karnataka government.
- Fine-tuned Google-MT5 base model for improved transliteration via transformer-based approaches, achieving 74% accuracy.
- Developed evaluation metrics for assessing translation quality and measuring improvements.

PUBLICATIONS & PATENTS

Accepted Conference, Journals & Workshop Papers

2026

- Vishwachetan D, Geetha Dayalan, Abhay Bhandarkar. “The Anatomy of a Resilient Phishing Detector: An In Depth Analysis of a Transformer’s Interpretability, Causality and Robustness.” *Frontiers in Computer Science journal* [Link] (Q2)
- Bhandarkar, A., S. Sridhar, N. Kiran Kumar, M. R., Rahul S.G., and T.M. Amirthalakshmi. PI-TFN: Enhancing Forecasting Robustness under Limited Historical Context via Physics-Informed Temporal Fusion. *Engineering Science and Technology, an International Journal (Elsevier)*, Vol. 81, 2026, Article 102483. [Link] (Q1)
- Abhay Bhandarkar, Vedanth Nanesha “VOUCH for Mathematical Reasoning: Counterfactual Contracts Cure the Self-Verification Pathology” *AI4Math Workshop, ICML 2026* [Link]
- A. Bhandarkar, R. Kaushik, V.N Balasubramanian “Pixels to Layers: Turning Generated Infographics into Editable Assets.” *AI4CC Workshop, CVPR 2026* [Oral Accept]. [Link]
- A. Bhandarkar, V.N Balasubramanian “Pixels to Layers: Turning Generated Infographics into Editable Assets.” *CVPR 2026 Demos* [Accepted]. [In Demos Session 1 List]
- Abhay Bhandarkar, Jamuna S Murthy, Amogh Kalasapura. “CAMMEL-ViT Net: Context Aware Memory Modules with Efficient Lightweight Vision Transformers for Lightweight Efficient Image Segmentation.” *Medical Image Understanding and Analysis (MIUA) 2026* [Oral Accept].
- Nikhil YN, Abhay Bhandarkar. “Helios: A Co-Designed Landscape-Aware Optimization System Bridging Serial Intelligence and GPU Parallelism.” *Accepted at EvoStar Applications 2026, Toulouse, France.* [Link]
- Abhay Bhandarkar, et al. “Causal Quantification of the Sensitivity-Reliability Trade-Off in Semantic XAI: Comparing Object-Aware (SAM) and Texture-Aware (SLIC) Segmentation.” *AAAI AI4Science Workshop 2026, Singapore.* [Link]

2025

- Abhay Bhandarkar, Vishwachetan D, Geetha B. “Augmenting URL-BERT based Phishing Detection with Favicon Metadata.” *International Conference on Computational Technologies for Research in Data Analytics: ICCTRDA 2025*.

2024

- Abhay Bhandarkar, Riddhi Rai, Sushma Bylaiah. “Advancements in Defense Surveillance: Integrating Deep Learning and Enhanced Materials in CubeSAT Systems.” *IEEE SPACE 2024*.
- Abhay Bhandarkar, Vishwachetan D. “Unmasking the AI Hand: A Machine Learning Approach to Deciphering Authorship.” *IEEE INOCON 2024*.

Top-Tier Submissions, Pre-prints & In Writing

2026

- Abhay Bhandarkar, Vineeth N Balasubramanian. “LEGO: Learning Expert-Guided Orchestration for Infographic Decomposition.” *Under Review at WACV 2027*.

2025

- Abhay Bhandarkar, Khushi, Gaurav Mishra et al. “Topic Modelling with BERTopic: Understanding the likings of humans across various LLMs via thorough analysis of LMSYS dataset.” (arXiv:2510.07557)

Patents

- **Patent (In process of grant):** Abhay Bhandarkar, Vineeth N Balasubramanian, Gongjie Qi, Bala Kumaravel, Microsoft Research
- **Patent Applied:** Abhay Bhandarkar, “JAM Algorithm for Probabilistic String Search”
- **Patent Applied:** Abhay Bhandarkar, “SIMHA: Sustainable Integrated Microbial and Hydrogen fuel cell Architecture”

AWARDS & ACHIEVEMENTS

Recognized as one of the EvoStar Outstanding Students of 2026 [Link] 2026
Selected as an ACM Summer Uplink Intern (16 selected out of 600+ applicants in India)[Link] 2025
Received \$50,000 from IBM in Compute Resources for research in space sustainability 2024/2025
1st place in CNS Hackathon (Solo participant) with MITRA AI Safety Framework
2nd prize in ACM Hackathon for CAMM-EL ViT Net medical imaging solution
2nd place among 800+ participants in Whackiest'24 by CodeRIT 2024
Qualified for Regionals in the Solving For India Hackathon by GFG X Google Cloud

REVIEWER & MENTORING EXPERIENCE

- **Emergency Reviewer**, CVIP, International Conference on Computer Vision & Image Processing (CVIP) Conference 2026.
- **Reviewer**, AI4Math Workshop, International Conference for Machine Learning (ICML) Conference 2026.
- **Reviewer**, PhilML Workshop, International Conference for Machine Learning (ICML) Conference 2026.
- **Reviewer**, AI4GOOD Workshop, International Conference for Machine Learning (ICML) Conference 2026.

- **Reviewer**, AI4CC Workshop, IEEE CVF Computer Vision and Pattern Recognition (**CVPR**) Conference 2026.
- **Reviewer**, IEEE Contemporary Computing Innovations Conference (CCIC) 2026.
- **Reviewer**, IEEE 1st International Conference on Global Symposium on Emerging and Communication Technologies (GSEACT 2026).
- **Research Mentor**, Mentoring 3 teams as a part of the IEEE CS IamPro student research internship (2026) in the domains of multimodality and applied computer vision.
- **Conference Mentor**, Guided juniors to publish impactful research at IEEE SPACE 2024 and 2025.

LEADERSHIP

IEEE Computer Society - RIT

2022 - 2025

Vice Chair and Chair

Bangalore, India

- Organized technical events and hackathons to bolster student engagement in computer science.
- Managed a committee of 45+ students to coordinate technical activities and facilitated workshops with industry professionals.

IEEE Student Team for Research & Innovation (ISTRI)

2024 - 2025

Founder

RIT, Bangalore, India

- Spearheaded a platform for real-world problem-solving research collaborations with IEEE-RIT.
- Created mentorship pathways connecting senior researchers with junior students.

S.T.A.R.D.U.ST – RIT

2023 - Aug 2023

Project Manager

Bangalore, India

- Contributed to flight computer design for Spaceport America Cup 2025.
- Led technical coordination between software and hardware teams, implementing documentation standards and version control.